METRIC

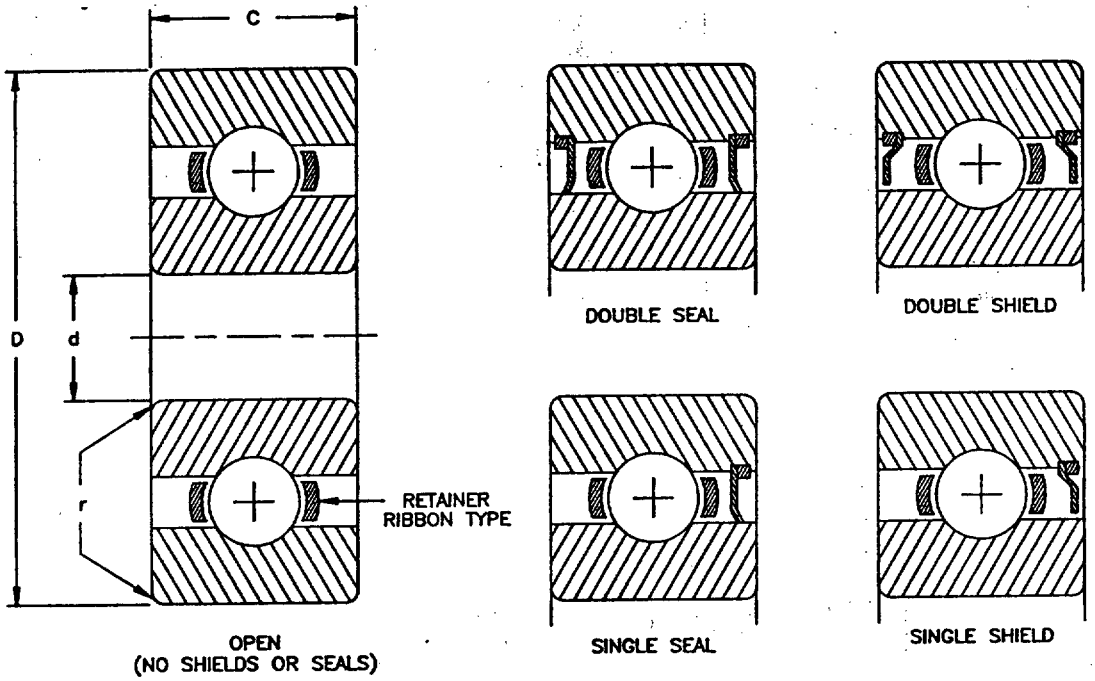
MIL-B-913/3
5 March 1993

MILITARY SPECIFICATION SHEET

BEARING, BALL, ANNULAR, FOR INSTRUMENTS AND PRECISION ROTATING COMPONENTS, DEEP GROOVE, UNFLANGED, CORROSION RESISTANT STEEL, ABEC 5P (METRIC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-B-913.



BEARING CONFIGURATIONS

AMSC N/A

FSC 3110

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REQUIREMENTS:

1. Design. All bearings described by this specification sheet shall be deep groove instrument bearings with an unflanged, rings flush, configuration.

2. Material. The ball and ring material for these bearings shall be corrosion resistant steel 440C (UNS S44004) conforming to QQ-S-763.

3. Tolerance class. The tolerance class shall be in accordance with the ABEC 5P tolerance tables of ANSI/AFBMA Standard 12.1. This tolerance class shall apply to all bearing sizes listed in Table I of this specification sheet.

4. Boundary dimensions. The boundary dimensions shall be as specified by the size code in Table I.

5. Retainer. The retainer shall be as specified by the part number designator (PN DES) in Table II.

6. Closures. Closures shall be as specified by the part number designator in Table III.

7. Radial internal clearance. The radial internal clearance shall be as specified by the part number designator in Table IV.

8. Calibration. Calibration shall be as specified by the part number designator in Table V.

9. Performance Test. The performance test shall be as specified by the part number designator in Table IV.

10. Lubrication.

10.1 Lubricant. The lubricant shall be in accordance with the specification specified by the part number designator in Table VI.

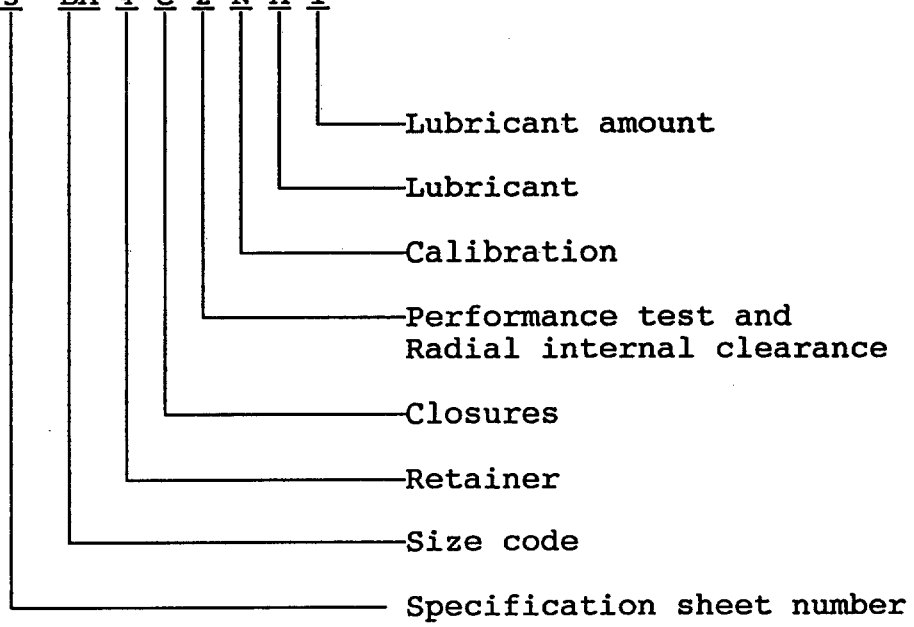
10.2 Lubricant amount. The amount of lubricant shall be as specified by the part number designator in Table VII.

10.3. Barrier coating. Barrier coating shall be applied to bearings as specified by the part number designator in Table VI.

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11. Part number. The part number consists of the following: prefix letter M; general specification number; number of this specification sheet; and alpha numeric code for size, retainer, closures, performance test and radial internal clearance, calibration, lubricant, and lubricant amount.

EXAMPLE: M 913/ 03 BA 4 C 2 N A 1



M913/03BA4C2NA1 indicates - Bore 2.0 mm; outside diameter 5.0 mm; width 1.5 mm; radius 0.08 mm; retainer, ribbon, loose clinched; closures, 2 shields; performance test starting torque, and radial internal clearance, 2 to 7 micrometers; no calibration; lubricant, MIL-L-6085; lubricant amount, manufacturer's standard

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TABLE I. Boundary dimensions (millimeters).

SIZE CODE	BORE	OD	WIDTH	RADIUS
	d	D	C	r1/
AA	1.5	5.0	2.0	0.15
BA	2.0	5.0	1.5	0.08
BB	2.0	5.0	2.3	0.08
BC	2.0	6.0	2.3	0.08
CA	2.5	6.0	1.8	0.08
CB	2.5	6.0	2.6	0.08
CC	2.5	7.0	2.5	0.15
DA	3.0	7.0	2.0	0.10
DB	3.0	7.0	3.0	0.10
DC	3.0	10.0	4.0	0.15
EA	4.0	9.0	2.5	0.10
EB	4.0	9.0	4.0	0.10
EC	4.0	13.0	5.0	0.15
ED	4.0	16.0	5.0	0.30
FA	5.0	11.0	3.0	0.15
FB	5.0	11.0	5.0	0.15
FC	5.0	16.0	5.0	0.30
FD	5.0	19.0	6.0	0.30
GA	6.0	13.0	3.5	0.15
GB	6.0	13.0	5.0	0.15
GC	6.0	19.0	6.0	0.30
HA	7.0	14.0	3.5	0.15
HB	7.0	14.0	5.0	0.15
HC	7.0	19.0	6.0	0.30
HD	7.0	22.0	7.0	0.30
HE	7.0	22.0	10.31	0.30
JA	8.0	19.0	6.0	0.20
JB	8.0	22.0	7.0	0.30
JC	8.0	22.0	10.31	0.30
JD	8.0	24.0	10.31	0.30
KA	9.0	20.0	6.0	0.30
KB	9.0	26.0	8.0	0.30
LA	10.0	26.0	8.0	0.50

1/ Maximum shaft or housing fillet radius that bearings corners will clear.

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TABLE II. Retainer.

PN DES	TYPE
1	MANUFACTURER'S STANDARD 1/
2	CROWN 2/
3	RIBBON, TIGHT CLINCHED 3/
4	RIBBON, LOOSE CLINCHED 3/
5	PTFE TUBE SEPARATOR 4/
6	NOT USED
7	PHENOLIC LAMINATE 5/, 7/
8	VACUUM IMPREGNATED PHENOLIC 6/, 7/
9	NON-POROUS, NON-METALLIC CROWN 7/

1/ One piece pressed corrosion resistant steel crown, or two piece pressed corrosion resistant steel ribbon.
2/ One piece pressed corrosion resistant steel.
3/ Two piece pressed corrosion resistant steel.
4/ PTFE (Polytetrfluoroethylene).
5/ Phenolic or other porous non-metallic materials.
6/ Phenolic or other porous non-metallic materials saturated with lubricant. Selection of this choice dictates choosing the manufacturer's standard amount of lubricant.
7/ Used for high speed applications.

TABLE III. Closures.

PN DES	NUMBER	TYPE
N	NONE	NONE
A	ONE	SHIELD 1/
C	TWO	SHIELD 1/
D	ONE	SEAL 2/
E	ONE	SEAL 3/
H	TWO	SEAL 2/
J	TWO	SEAL 3/

1/ Corrosion resistant steel conforming to ASTM-A-580, Condition A and ASTM-A-313, Condition C or QQ-S-766.
2/ Synthetic rubber.
3/ Glass fiber (Polytetrafluoroethylene) or other inert fiber.

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TABLE IV. Performance Test — Radial internal clearance

PN DES	TYPE	RANGES (micrometers)
1	MANUFACTURER'S STANDARD	2 to 7
2	STARTING TORQUE 1/	
3	MANUFACTURER'S STANDARD	5 to 12
4	STARTING TORQUE 1/	
5	MANUFACTURER'S STANDARD	12 to 20
6	STARTING TORQUE 1/	
7	MANUFACTURER'S STANDARD	20 to 28
8	STARTING TORQUE 1/	

1/ Starting torque limits from table III of MIL-B-913.

TABLE V. Calibration of bore and outside diameter (micrometers).

PN DES	BORE INCREMENTS	OD INCREMENTS
N	NO CALIBRATION	NO CALIBRATION
A	NO CALIBRATION	2.50
B	NO CALIBRATION	1.25
C	2.50	NO CALIBRATION
D	2.50	2.50
E	2.50	1.25
F	1.25	NO CALIBRATION
G	1.25	2.50
H	1.25	1.25

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TABLE VI. Lubricant.

PN DES	SPECIFICATION
P	PRESERVATIVE 1/
A	MIL-L-6085
B	MIL-L-6085 2/
C	MIL-L-81846
D	MIL-L-81846 2/
E	MIL-G-23827
F	MIL-G-81322
G	MIL-G-81937
H	MIL-G-83261
J	MIL-S-81087
K	MIL-S-81087 2/

1/ PN DES "P" shall be used only with PN DES "P" in TABLE VII.
2/ With barrier coat.

TABLE VII. Lubricant amount.

PN DES	QUANTITY
P	PRESERVATIVE AMOUNT AS REQUIRED BY MIL-P-197
1	MANUFACTURER'S STANDARD 1/
2	OIL PER TABLE II of MIL-B-913
3	15% GREASE PER TABLE II of MIL-B-913
4	25% GREASE PER TABLE II of MIL-B-913
5	35% GREASE PER TABLE II of MIL-B-913
6	45% GREASE PER TABLE II of MIL-B-913

1/ Grease, fill to minimum 25%, maximum 40% of bearing void. Oil, immerse and then allow excess to drip off. The standard quantity of oil varies with each bearing size.

Custodians:
Army - AT
Navy - AS
Air Force - 99

Review activities:
Navy - SH
DLA - IS

User activities:
Navy - MC
Air Force - 84

Preparing activity:
Navy - AS

Agent:
DLS - IS

(Project 3110-0813-03)

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